GCNFormNet: branched graph-transformer architecture for EEG-based emotion recognition

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AGENDA

Outline

Introduction

1. **Why emotion recognition?**

   Affective computing underpins adaptive e-learning, empathetic healthcare robots and real-time mental-health monitoring.

2. **Why EEG?**

   Behavioural cues (face, speech) can be consciously masked. EEG traces the neural processes behind an emotion with high temporal resolution.

3. **The gap**

   Most state-of-the-art pipelines depend on heavy multi-stage preprocessing and handcrafted features, obscuring what the architecture itself contributes.

4. **This paper**

   An end-to-end deep architecture that learns spatio-temporal structure directly from minimally processed EEG.

KEY POINTS

- EEG-based emotion recognition
- Minimal, standardised preprocessing
- Joint spatial + temporal learning
- Dual-branch: GCN + Transformer
- Subject-independent (LOSO) evaluation

Background

Established architectures

- CNNs on topographic maps — Bashivan et al. (2015)
- EEGNet — compact depthwise / separable CNN (2018)
- Shallow & Deep ConvNet — standard CNN baselines
- TSception — multi-scale temporal + asymmetry
- Vision Transformers and hybrid CNN–Transformers (MSDCGNet)
- Hybrid CNN–LSTM / GRU–RNN models

Recurring problems

- Manual feature extraction — Differential Entropy, PSD, wavelet transforms
- Topology-dependent 2D/3D/4D feature maps
- Inconsistent protocols: LOTO reports 80–95%, LOSO only 50–60%
- Seven different valence thresholds used for DEAP alone
- Cross-dataset accuracy collapses to near-chance (domain shift)

THE GAP

Problem Statement

1 **Heavy preprocessing**

Multi-stage filtering and transformation pipelines mask the true capability of the underlying architecture.

2 **Handcrafted features**

DE / PSD / wavelet features embed prior assumptions and are costly to compute per band.

3 **Weak subject independence**

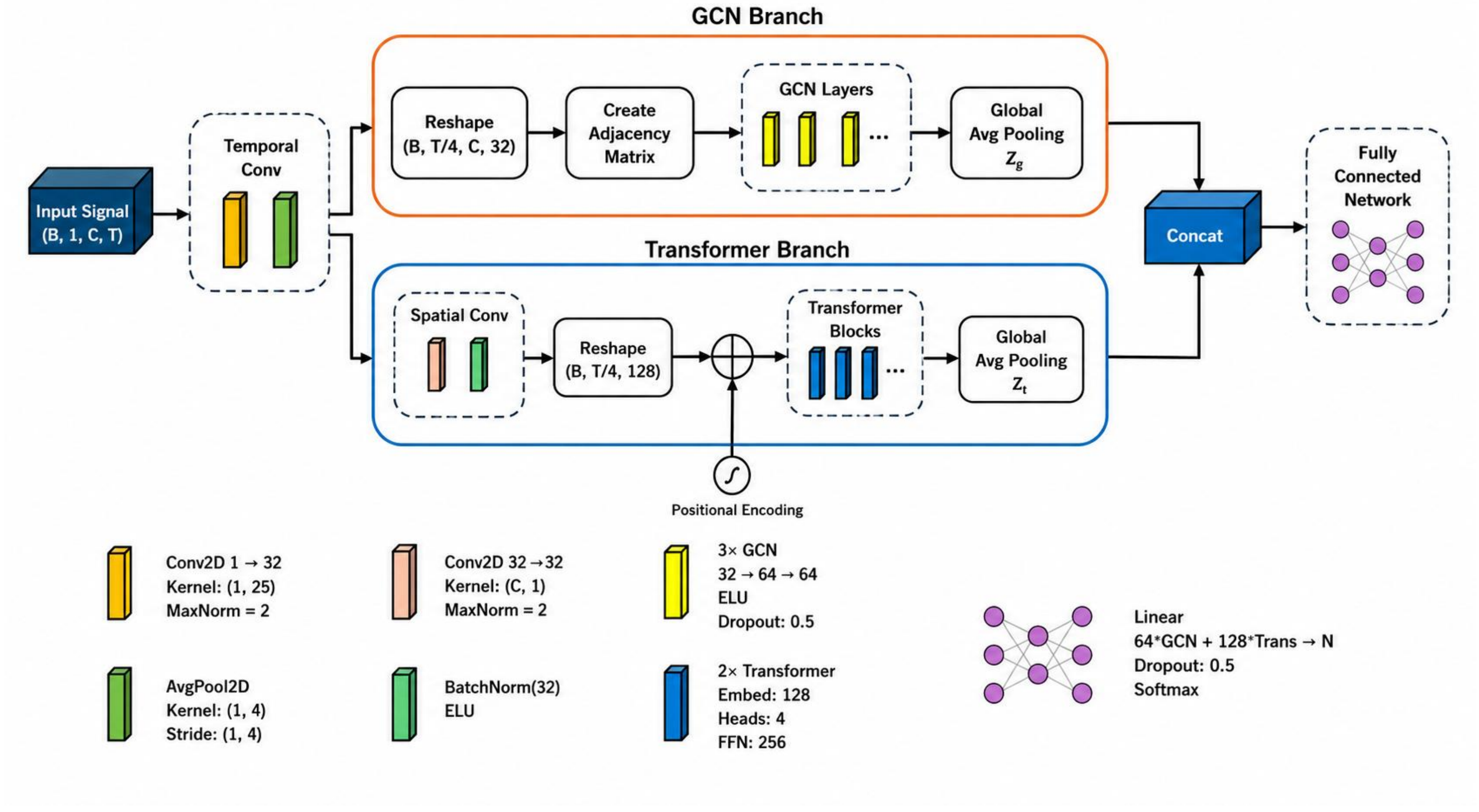
Under LOSO, complex models often fail to beat a trivial majority-class predictor.

4 **Incomparable protocols**

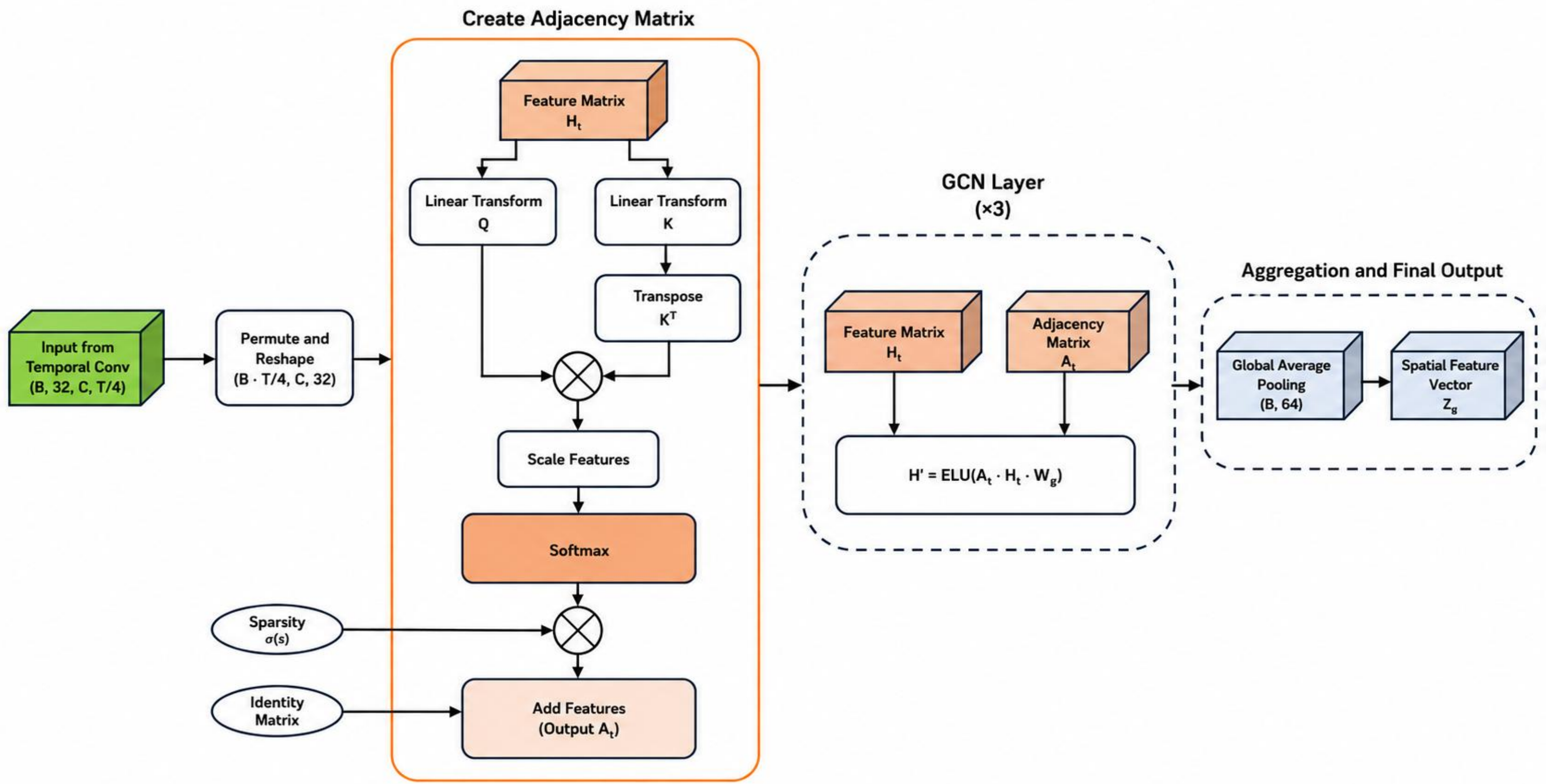
Different splits, thresholds and metrics make published numbers impossible to compare.

OBJECTIVE Design a model that learns robust spatio-temporal representations directly from minimally processed EEG — and validate it under a unified, subject-independent protocol.

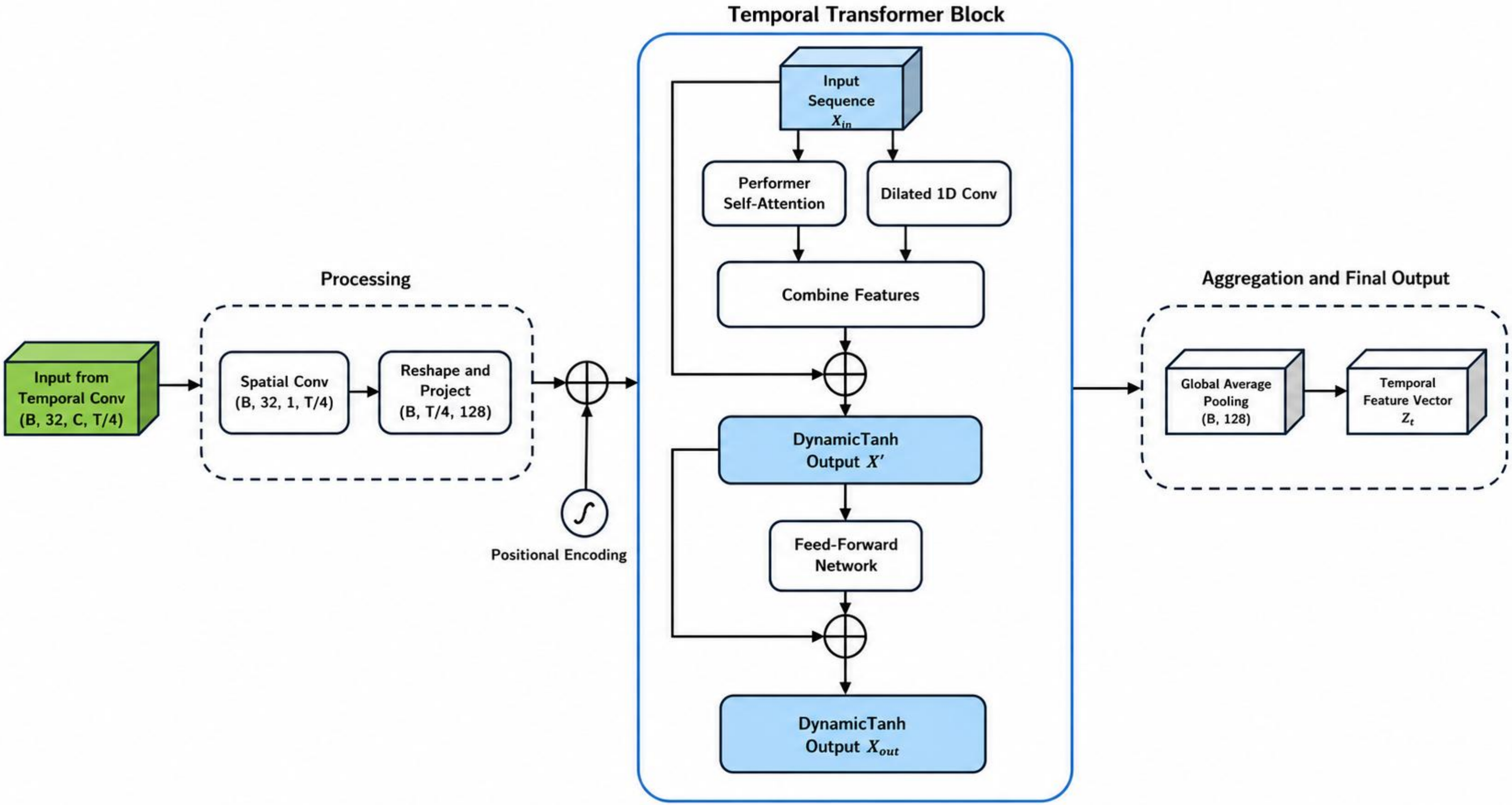
GCNFormNet Architecture Overview



Spatial Graph Module: Adjacency Matrix Construction & GCN Layers



Temporal Module: Temporal Transformer Block



Brief Representation of the Work

1 — Input
Minimally processed EEG 128 Hz, band-pass, 4 s windows

→

2 — Temporal features
Shared temporal convolution over raw channels

→

3 — Dual-branch encoding
GCN learns spatial relations
Transformer learns temporal ones

→

4 — Feature fusion
Concatenate Z^g and Z_t with dropout

→

5 — Classification
Dense layer + softmax over emotion classes

Main contributions

- **Dynamic adjacency matrix**
  Inter-channel graph re-built at every time step — no anatomical prior.
- **GCN + Transformer hybrid**
  Spatial and temporal dependencies modelled in parallel branches.
- **DynamicTanh over LayerNorm**
  Normalisation-free training; higher val. accuracy and faster throughput.
- **Minimal preprocessing**
  No DE / PSD / wavelet features — the architecture does the work.
- **Unified EEGain + LOSO evaluation**
  Four benchmarks, one protocol, reproducible comparison.

CORE EQUATIONS

Mathematical Formulation

1 · DYNAMIC ADJACENCY MATRIX

$$A_t = \sigma(s) \cdot \mathrm{Softmax}\left(\frac{(H_t W_q)(H_t W_k)^T}{\tau}\right) + I$$

τ = temperature, s = learnable sparsity, I adds self-loops. Row-normalised by softmax.

2 · GRAPH CONVOLUTION (× 3 LAYERS)

$$H_t^{(l+1)} = \mathrm{ELU}\left(A_t H_t^{(l)} W_g^{(l)}\right)$$

Hidden dimension 64; Z^g = mean over channels, then over time steps.

3 · TEMPORAL TRANSFORMER BLOCK

$$X' = \mathrm{DyT}_1(X_{\mathrm{in}} + \mathrm{Performer}(X_{\mathrm{in}}) + \mathrm{DilatedConv}(X_{\mathrm{in}}))$$
$$H_{\mathrm{out}} = \mathrm{DyT}_2(X' + \mathrm{FFN}(X'))$$

DynamicTanh replaces LayerNorm; Performer gives linear-cost attention.

4 · FEATURE FUSION

$$Z_{\mathrm{comb}} = [Z_g;\, Z_t]$$

Spatial and temporal vectors concatenated (64 + 128).

5 · CLASSIFICATION

$$\hat{y} = \mathrm{Softmax}(W_c \cdot \mathrm{Dropout}(Z_{\mathrm{comb}}) + b_c)$$

Dropout 0.5; cross-entropy with 0.01 label smoothing.

NOTATION

B batch · C channels · T time steps · F filters
$X \in \mathbb{R}$^(B×1×C×T) input; $H_t \in \mathbb{R}$^(C×F) per-step features
A_t time-varying adjacency; Z^g, Z_t branch outputs
σ sigmoid; τ = 0.1; s_{init} = 0.5 (defaults)

Experimental Setup

Four EEGain benchmark datasets

Dataset	Subjects	Channels	Task
DEAP	32	32	Binary valence / arousal
DREAMER	23	14	Binary valence / arousal
SEED	15	62	3-class categorical
SEED-IV	15	62	4-class categorical

Standardised EEGain pipeline: 128 Hz resampling, band-pass / notch filtering, 4-second windows.

Baselines: EEGNet · ShallowConvNet · DeepConvNet · TSception · MSC-Attention-LSTM · Trivial (majority-class) baseline.

Training configuration

LOSO
Leave-One-Subject-Out

200
Training epochs

32
Batch size

0.001
Adam learning rate

0.5
Dropout rate

2025
Fixed random seed

Results & Discussion

Dataset	Task	Best Baseline (Acc.)	GCNFormNet (Acc.)	GCNFormNet F1	Significance
DEAP	Valence	Trivial — 0.57	0.50 ± 0.09	0.55 ± 0.18	Underperforms Trivial (p<0.01)
DEAP	Arousal	Trivial — 0.59	0.52 ± 0.11	0.51 ± 0.20	Underperforms Trivial (p<0.05)
DREAMER	Valence	MSC-Attn-LSTM — 0.64	0.57 ± 0.07	0.40 ± 0.15	Underperforms MSC-LSTM (p<0.001)
DREAMER	Arousal	Trivial — 0.52	0.48 ± 0.08	0.45 ± 0.15	No significant difference
SEED	3-class	DeepConvNet — 0.55	**0.54 ± 0.07**	**0.52 ± 0.07**	**Outperforms 4/6 baselines**
SEED-IV	4-class	DeepConvNet — 0.45	0.46 ± 0.08	0.43 ± 0.10	**Best overall; outperforms 4/6**

Per-dataset outcome

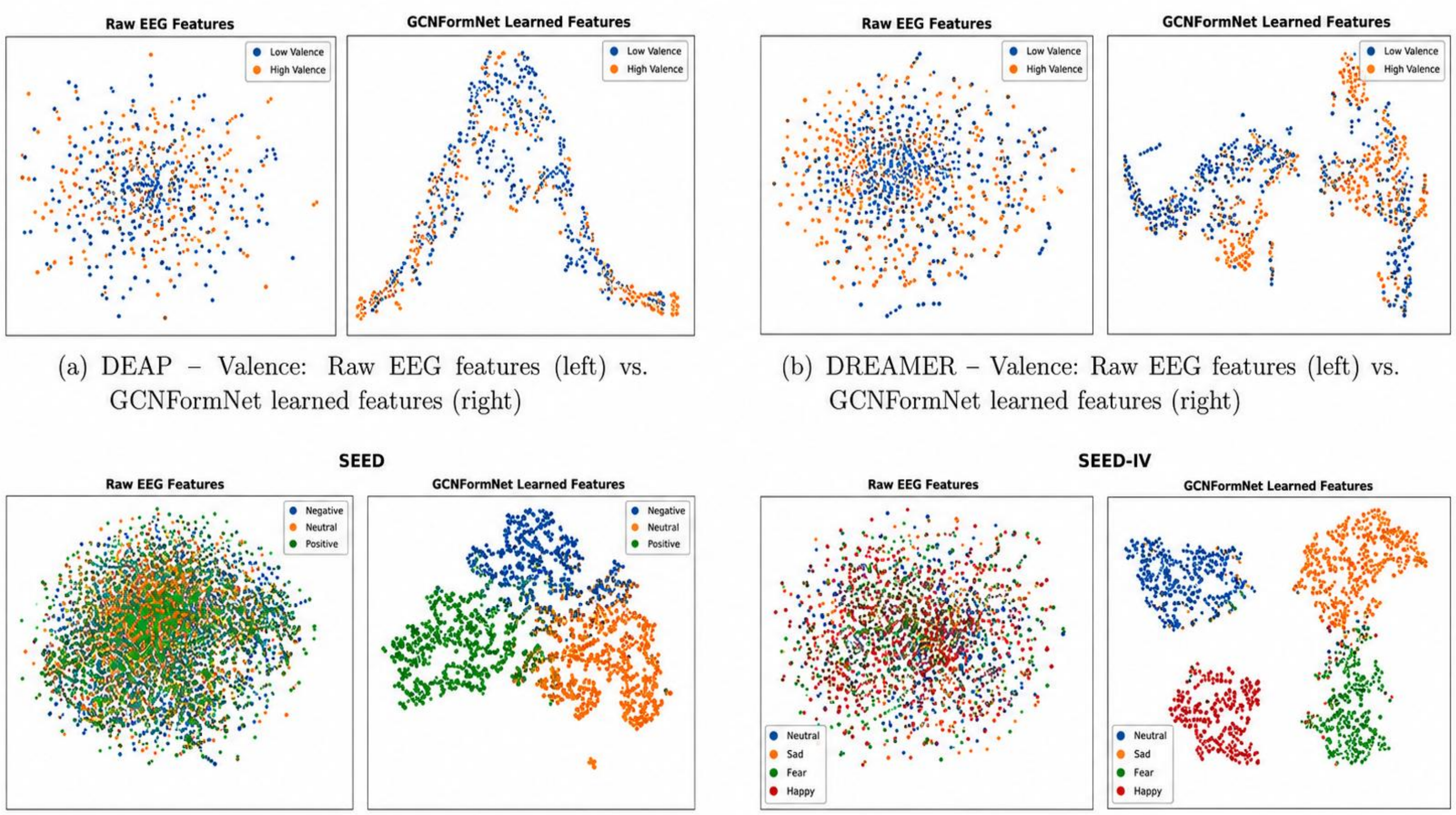
- **SEED-IV** — best of all models
- **SEED** — second, matches DeepConvNet
- **DREAMER** — competitive
- **DEAP** — below trivial baseline

Main findings

- Highest SEED-IV accuracy; gains over EEGNet, ShallowConvNet, TSception and the trivial baseline are statistically significant (Welch's t-test).
- Strongest on 62-channel categorical tasks; binary tasks show narrow, mostly non-significant gaps.

Interpretability: What the Model Learned

Learned feature separability (t-SNE) and dynamic functional connectivity — discovered purely from data



(a) DEAP – Valence: Raw EEG features (left) vs. GCNFormNet learned features (right)

(b) DREAMER – Valence: Raw EEG features (left) vs. GCNFormNet learned features (right)

(c) SEED – 3-class: Raw EEG features (left) vs. GCNFormNet learned features (right)

(d) SEED-IV – 4-class: Raw EEG features (left) vs. GCNFormNet learned features (right)

Fig. : t-SNE, raw vs. learned features

DREAMER — Valence

Clear hemispheric block structure. Low valence relies on inter-hemispheric coupling; high valence shows stronger right-frontal intra-hemispheric links (AF4, F8, F4) - matching frontal asymmetry theory.

DEAP — Valence

High valence: stronger temporal-frontal (T8-Fp1) & occipital inter-hemispheric coupling (O1-O2). Low valence: more diffuse connectivity.

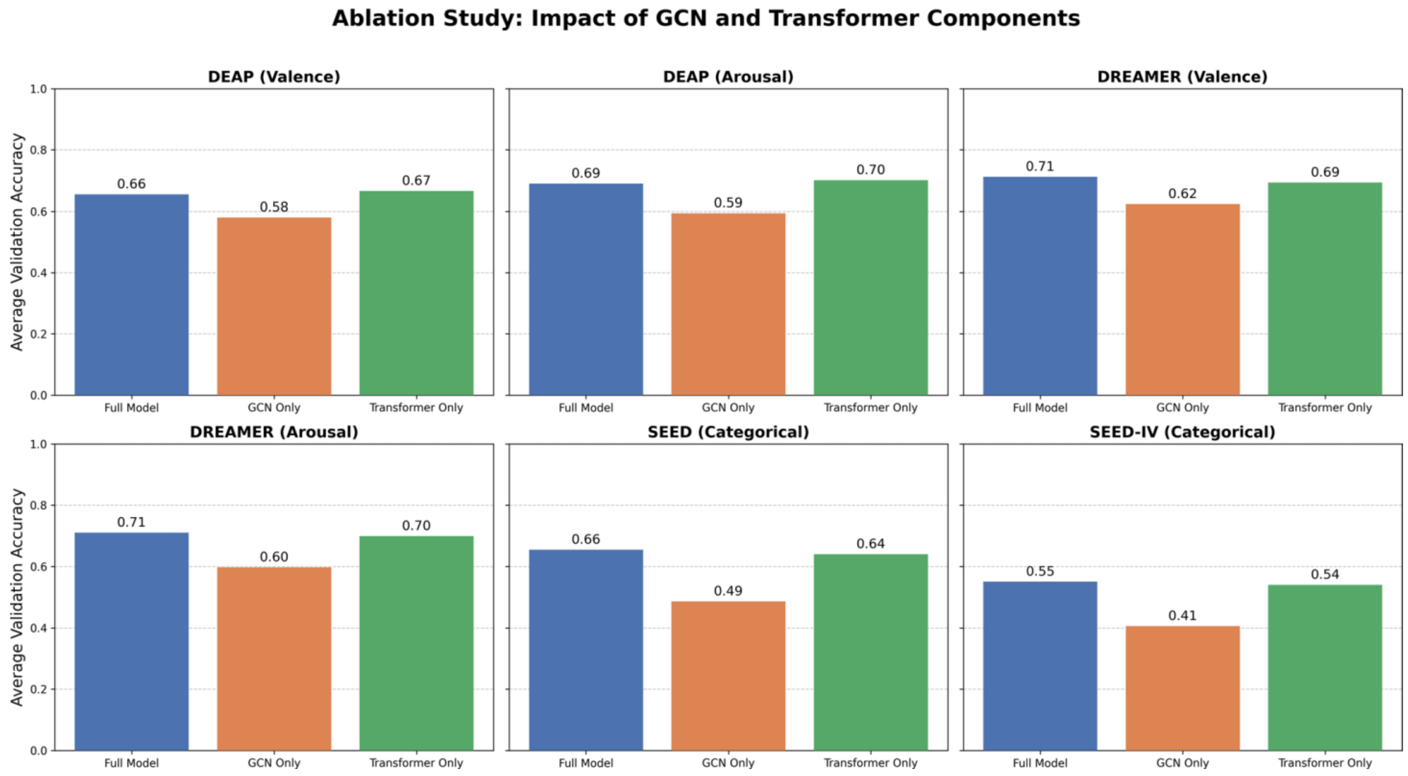
SEED — 3-class emotion

All classes show prefrontal-dominant connectivity. Neutral has the highest prefrontal self-connectivity, then Positive, then Negative - model's key axis is engagement vs. Neutral baseline.

SEED-IV — 4-class emotion

Distinct region profile per emotion: Sad = highest prefrontal self-connectivity (rumination-like); Happy = distributed, posterior-engaged; Fear = anterior-focused; Fear-Happy overlap reflects shared high arousal.

Ablation Study & Key Takeaways



Dataset	Task	Model	Accuracy	F1	F1 Weighted
DEAP	Valence	**Full Model**	0.50 ± 0.09	0.55 ± 0.18	0.44 ± 0.09
		without GCN	0.50 ± 0.06	0.55 ± 0.13	$\mathbf{0.46 \pm 0.08}$
		without Transformer	$\mathbf{0.53 \pm 0.08}$	$\mathbf{0.63 \pm 0.16}$	0.45 ± 0.10
	Arousal	**Full Model**	0.52 ± 0.11	0.51 ± 0.20	$\mathbf{0.50 \pm 0.12}$
		without GCN	0.51 ± 0.10	0.55 ± 0.16	$\mathbf{0.50 \pm 0.11}$
		without Transformer	$\mathbf{0.56 \pm 0.14}$	$\mathbf{0.67 \pm 0.15}$	0.45 ± 0.16
DREAMER	Valence	**Full Model**	$\mathbf{0.57 \pm 0.07}$	$\mathbf{0.40 \pm 0.15}$	$\mathbf{0.54 \pm 0.08}$
		without GCN	0.56 ± 0.07	0.38 ± 0.09	0.54 ± 0.07
		without Transformer	$\mathbf{0.57 \pm 0.08}$	0.20 ± 0.16	0.49 ± 0.11
	Arousal	**Full Model**	$\mathbf{0.48 \pm 0.08}$	$\mathbf{0.45 \pm 0.15}$	$\mathbf{0.46 \pm 0.08}$
		without GCN	0.47 ± 0.06	0.43 ± 0.15	$\mathbf{0.46 \pm 0.07}$
		without Transformer	$\mathbf{0.48 \pm 0.08}$	0.39 ± 0.16	$\mathbf{0.46 \pm 0.09}$
SEED	Categorical	**Full Model**	$\mathbf{0.53 \pm 0.05}$	$\mathbf{0.52 \pm 0.06}$	$\mathbf{0.52 \pm 0.06}$
		without GCN	0.52 ± 0.05	0.51 ± 0.06	0.51 ± 0.06
		without Transformer	0.44 ± 0.06	0.40 ± 0.06	0.40 ± 0.06
SEED-IV	Categorical	**Full Model**	$\mathbf{0.46 \pm 0.08}$	$\mathbf{0.43 \pm 0.10}$	$\mathbf{0.44 \pm 0.10}$
		without GCN	0.45 ± 0.08	0.42 ± 0.08	0.42 ± 0.08
		without Transformer	0.37 ± 0.06	0.26 ± 0.07	0.29 ± 0.07

Ablation findings

- Transformer branch = consistent backbone; GCN-only is weakest across all datasets
- Full hybrid model wins on DREAMER, SEED, SEED-IV → confirms synergy between branches
- DEAP exception: GCN-only edges out full model on F1 — GCN overfits to subject-specific patterns on this lower-density, 32-channel, high-variability dataset

CRITICAL APPRAISAL

Advantages & Limitations

Advantages

- **End-to-end learning**

  No handcrafted spectral features required.

- **Dynamic graph learning**

  Connectivity inferred per time step, not fixed by topology.

- **Explainability**

  Adjacency maps and t-SNE give a readable account of the decision.

- **Efficient design**

  Performer attention is linear-cost; DyT trains faster than LayerNorm.

- **Stable under LOSO**

  Lower variance across held-out subjects than most baselines.

Limitations

- **Weak on DEAP**

  The trivial majority-class baseline beats the full model on valence ($p < 0.01$).

- **Channel-count dependence**

  On 32 or 14 electrodes the dynamic graph captures spurious correlations.

- **Inter-subject variability**

  Per-fold accuracy ranges from near-chance to about 55% on SEED-IV.

- **Absolute accuracy is modest**

  0.46 on a 4-class task — realistic for LOSO, but far from deployable.

- **Needs domain adaptation**

  Minimal preprocessing alone cannot remove subject-specific neural patterns.

CONCLUSION

A hybrid architecture that earns its accuracy from design, not preprocessing

1 **Dual-branch design**

GCN captures dynamic spatial structure; the Performer-based Transformer captures long-range temporal dynamics.

2 **Dynamic adjacency**

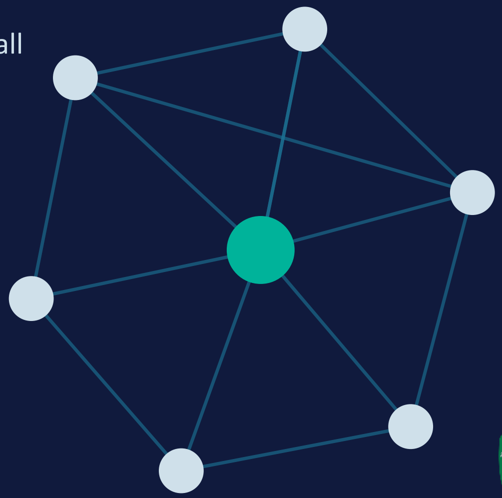
Context-specific inter-channel graphs learned per time step, without anatomical priors.

3 **DyT validated**

DynamicTanh replaces LayerNorm with higher validation accuracy and faster training and inference.

4 **Best on SEED-IV**

0.46 accuracy under LOSO — the strongest result among all compared models.



FUTURE WORK

Adaptive branch weighting via gating or attention, plus evaluation across all six EEGain benchmark datasets.

References

1. Raghav, A., & Indolia, S. (2026). GCNFormNet: branched graph-transformer architecture for EEG-based emotion recognition. Cognitive Neurodynamics, 20:66. https://doi.org/10.1007/s11571-026-10437-z

Thank You

Questions & Answers

Raghav, A. & Indolia, S. (2026). GCNFormNet: branched graph-transformer architecture for EEG-based emotion recognition. Cognitive Neurodynamics, 20:66.